

Gyuyeon Kim

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EDUCATION

Korea Advanced Institute of Science and Technology

March 2025 – Ongoing

Ph.D., School of Computing

Advisor: Jun Han

Yonsei University

March 2023 – February 2025

M.S., Electrical and Electronic Engineering

Advisor: Jun Han

Yonsei University

Mar 2017 – February 2023

B.S., Electrical and Electronic Engineering

GPA: 4.02/4.3

RESEARCH INTEREST

My research focuses on developing novel applications in mobile and sensing systems by leveraging physics-based artificial intelligence to enable new forms of perception and inference that were previously unattainable.

PUBLICATIONS

- **Gyuyeon Kim**, Sehoon Lim, Jaewoo Son, Soundarya Ramesh, and Jun Han, **AcousTag: Batteryless 3D-Printed Acoustic Tag for Smart Speaker-based Event Monitoring**, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), 2026.
- **Gyuyeon Kim**, Jaewoo Son, Sungmin Park, and Jun Han, **AutoScale: Vision-Based Overweight Truck Detection Utilizing A Following Autonomous Vehicle**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
- Bangjie Sun, Kanav Sabharwal, **Gyuyeon Kim**, Mun Choon Chan, and Jun Han, **Testing Masks and Air Filters with Your Smartphones**, ACM Conference on Embedded Networked Sensor Systems (SenSys), 2023.
- **Gyuyeon Kim***, Jonghyuk Yun*, Soundarya Ramesh and Jun Han, **RampScope: Ramp-Level Localization of Shared Mobility Devices Using Sidewalk Ramps**, ACM International Workshop on Mobile Computing Systems and Applications (HotMobile), 2023.

POSTERS AND DEMOS IN PROCEEDINGS

- **Gyuyeon Kim**, Hyunwoo Kim, and Jun Han, **Poster: Towards Acoustic-Based Tagless Object Tracking with Smartwatches**, ACM International Conference on Mobile Systems, Applications, and Services (MobiSys), 2024.
- **Gyuyeon Kim***, Jonghyuk Yun*, Soundarya Ramesh and Jun Han, **RampScope: Ramp-Level Localization of Shared Mobility Devices Using Sidewalk Ramps**, ACM International Workshop on Mobile Computing Systems and Applications (HotMobile), 2023.

AWARD AND HONORS

Student Travel Grant

ACM International Conference on Mobile Systems, Applications, and Services (MobiSys '24)

2024

Best Poster Award

ACM Workshop on Mobile Computing Systems and Applications (HotMobile '23)

2023

Student Travel Grant

ACM Conference on Embedded Networked Sensor Systems (SenSys '23)

2023

Presidential Science Scholarship
Korea Student Aid Foundation (KOSAF)

2017-2023

High Honor Student Awards
Yonsei University

2023

Honor Student Awards
Yonsei University

2021

RESEARCH EXPERIENCE

Research Assistant
Cyber-Physical Systems and Security (CyPhy) Lab, KAIST

Daejeon, Republic of Korea
Feb 2024 - Feb 2025

Research Intern
Cyber-Physical Systems and Security (CyPhy) Lab, Yonsei University

Seoul, Republic of Korea
Jan 2022 - Feb 2023

Military Service
Korean Augmentation To the United States Army (KATUSA)

Daegu, Republic of Korea
Aug 2019 – Mar 2021

TEACHING EXPERIENCE

Teaching Assistant

- [IS59900] Introductory Special Topics in Security and Privacy, KAIST, Fall 2025
- [CS30401] Introduction to Computer Networks, KAIST, Spring 2025
- [EEE3450] Communication Network, Yonsei University, Fall 2023
- [ENG1108] Engineering Information Processing, Yonsei University, Fall 2021

Project Mentor

- [CS442] Mobile Computing and Applications, KAIST, Spring 2024

PROFESSIONAL ACTIVITIES

External Reviewer

- ACM MobiSys (2023, 2024, 2025, 2026), ACM MobiCom (2025, 2026), ACM SenSys (2022, 2023, 2024, 2026), ACM IMWUT (2026), ACM/IEEE IPSN (2024)
- IEEE Security and Privacy (Oakland) (2025, 2026), USENIX Security (2024)
- ACM UIST (2025, 2026)
- ACM HotMobile (2023, 2024, 2025), ACM WiSec (2024), IEEE IoTDI (2023), IEEE VT (2023), ACM BuildSys (2022)

SKILLS

Programming Languages: Python, MATLAB, C/C++ , Java, Kotlin

Framework: PyTorch, Tensorflow, Simulink, Unreal Engine, Android

Tools: Git, Docker, Adobe Illustrator

Languages: English (Fluent), Korean (Native)